

CHAPTER

ELECTROSTATICS

Electrostatics

→ charge at rest

The charge on a body whose magnitude is unity is called unit charge.

point charge → no → l, b, h

* flow of e^- = charge

→ $e^- + e^- + e^- \dots = nq$

→ excess e^- = -ve

→ deficiency e^- = +ve

There are two types of charge distribution

→ discrete charge distribution

→ continuous charge distribution

Discrete Charge Distribution

- if charges are countable then called D.C.D
- In D.C.D, we assume there is no force b/w charges in a system

Continuous Charge Distribution

- if charges are uncountable
- In C.C.D, there are 8 ways of distribution

$\Delta \rightarrow$ density

i) Surface Charge Distribution (S.D.C)

ii) Volume Charge Distribution (V.C.D)

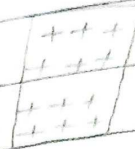
iii) Linear charge Distribution (L.D.C)

Surface Density of Charge

Surface OR Distribution of Charge

Charge per unit area is called Surface density of charge.

$$\text{SDC} = \frac{\text{Charge}}{\text{Area}}$$



$$b = dy = 1\text{m}$$

$$c = dx = 1\text{m}$$

$$\text{Area} = dx dy = 1\text{m}^2$$

$$\sigma = \frac{dq}{da}$$

$$dq = \sigma da \quad \dots (1)$$

if area is not given in units

if area is not given in units then, first we divide the area into small unit areas $da_1, da_2, da_3, \dots$



Integrating into eq (1)

$$\int dq = \sigma \int da$$

$$dx \cdot c \neq b = 1\text{m}$$

$$da \neq \text{m}^2$$

$$q = \sigma \int da$$

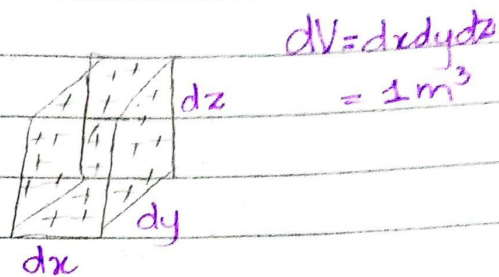
$$q = \sigma \iint_s dx dy$$

$$q = \sigma \iint_s dx dy$$

Volume Density Of Charge

OR

Volume Distribution Of Charge



The charge inside the surface is 'volume density of charge.' OR Charge per unit volume is called 'volume density of charge.'

$$\text{VDC} = \frac{\text{Charge}}{\text{Volume}}$$

$$\rho = \frac{dq}{dV}$$

$$dq = \rho dV \quad \dots\dots (2)$$

if volume is not in units
 $dV_1, dV_2, \dots\dots$

integrating eq (2)

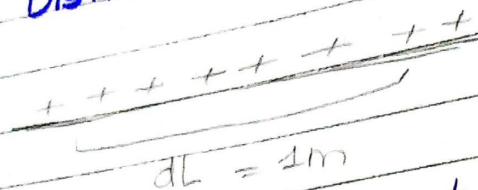
$$\int dq = \rho \int dV$$

$$q = \rho \int dV$$

$$q = \rho \iiint dz dy dx$$

$$q = \rho \iiint dx dy dz$$

Linear Density OF Charge OR Linear Distribution OF Charge



Charge on the line is known as linear density of Charge. or charge per unit length is called linear density of charge.

$$LDC = \frac{\text{Charge}}{\text{length}}$$

$$\lambda = \frac{dq}{dL}$$

$$dq = \lambda dL \dots (3)$$

if length in not unit

integrating eq (3)

$$\int dq = \lambda \int dL$$

$$dq_L = \lambda \int_L dL$$

$$dq_L = \lambda \int_L dL$$

COLUMB'S LAW



The force of attraction or repulsion b/w two similar or dissimilar charges placed at a distance apart from each other:

- force is directly proportional to the product of two charges (q_1 & q_2).
- force is inversely proportional to the square of the distance b/w two charges q_1 and q_2 .

Mathematically,

$$F \propto q_1 q_2$$

$$F \propto 1/r^2$$

combining (i) and (ii)

$$F \propto \frac{q_1 q_2}{r^2}$$

$$\therefore K = \frac{1}{4\pi\epsilon_0}$$

$$F = K \frac{q_1 q_2}{r^2}$$

$$F = \frac{1}{4\pi\epsilon_0} \cdot \frac{q_1 q_2}{r^2}$$

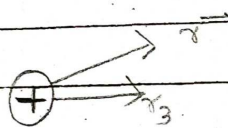
Vector Form

$$\vec{F} = |\vec{F}| \hat{r}$$

$$\vec{F}_{12} = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{r^2} \hat{r}$$

Force Due to N-Charges OR

Force b/w point charges or System Of Charge



Consider N-point charges $q_1, q_2, q_3, \dots, q_n$ placed at a distance of $r_1, r_2, r_3, \dots, r_n$ from the charge q_0 as shown in figure.

Charges q_1 to q_n exert the force on q_0 which is given by coulombs law;

$$\vec{F}_1 = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_0}{r_1^2} \hat{r}_1$$

$$\vec{F}_2 = \frac{1}{4\pi\epsilon_0} \frac{q_2 q_0}{r_2^2} \hat{r}_2$$

$$\vec{F}_n = \frac{1}{4\pi\epsilon_0} \frac{q_n q_0}{r_n^2} \hat{r}_n$$

The Total Force on charge q_0 is the sum of all the forces

$$\vec{F}_{\text{total}} = \vec{F}_1 + \vec{F}_2 + \dots + \vec{F}_n$$

$$\vec{F}_{\text{total}} = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_0}{r_1^2} \hat{r}_1 + \frac{1}{4\pi\epsilon_0} \frac{q_2 q_0}{r_2^2} \hat{r}_2 + \dots + \frac{1}{4\pi\epsilon_0} \frac{q_n q_0}{r_n^2} \hat{r}_n$$

$$\vec{F}_{\text{total}} = \frac{q_0}{4\pi\epsilon_0} \left[\frac{q_1}{r_1^2} \hat{r}_1 + \frac{q_2}{r_2^2} \hat{r}_2 + \dots + \frac{q_n}{r_n^2} \hat{r}_n \right]$$

$$\vec{F}_{\text{total}} = \frac{q_0}{4\pi\epsilon_0} \left[\text{Sum of } \frac{q_i \hat{r}_i}{r_i^2} \right]$$

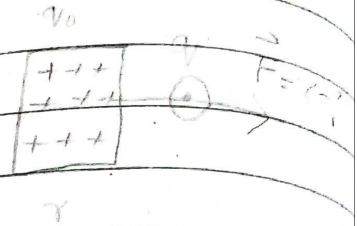
$$\vec{F}_{\text{total}} = \frac{q_0}{4\pi\epsilon_0} \sum_{n=1}^n \frac{q_n}{r_n^2} \hat{r}_n$$

This gives the total force due to point charge or system of charges.

Force Due to Surface Density Of Charge

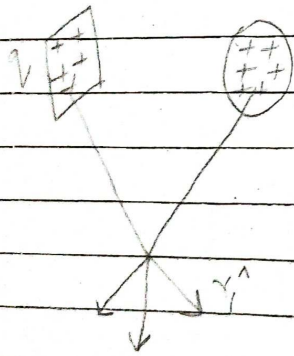
$$F = \frac{1}{4\pi\epsilon_0} \frac{q_3 q_2}{r^2} \hat{r}$$

$$q_3 = \sigma \iint dx dy$$

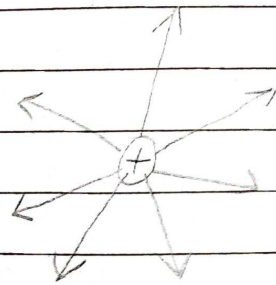


Force Due to Surface Density Of Charge and Volume Density Of Charge

→ On a Point Charge



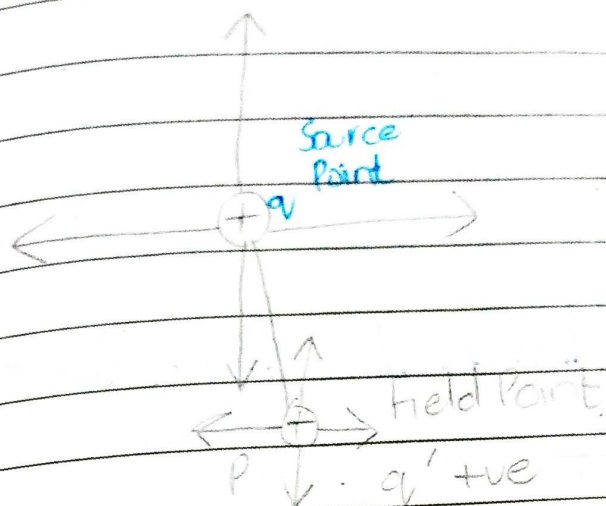
$$\vec{r} = \vec{F}_s + \vec{F}_v$$



The space around the charge within which its effect can be felt. It is represented by arrows due to positive charge, arrows will be outward (**Divergence**) and due to negative charge, arrows will be inward (**Convergence**).

Electric field always directed from +ve to -ve

Electric Field Intensity



Consider a +ve charge q placed at point 'O'. The point O at which charge q is placed is known as **source point**.

Let a unit positive charge q' be placed at point 'P'. The point at which q' is placed is known as **field point**.

The force with which q repels q' is known as **electric field intensity**.

$E \cdot F \cdot I$ = force per unit charge

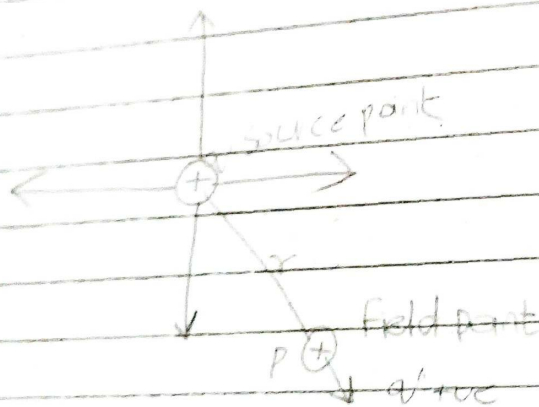
$$E \cdot F \cdot I = \frac{\text{force}}{\text{charge}}$$

$$E = \frac{F}{q} \quad \text{eq 1}$$

electric field intensity is a vector quantity.
its direction is same as the direction of force.

$$\text{unit} = \frac{N}{C}$$

Electric Field Intensity Due to a Charge



same from (1-5) written in electric field intensity previously

6) To determine the force b/w q and q' which is given by columb's law

$$\vec{F} = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{r^2} \hat{r}$$

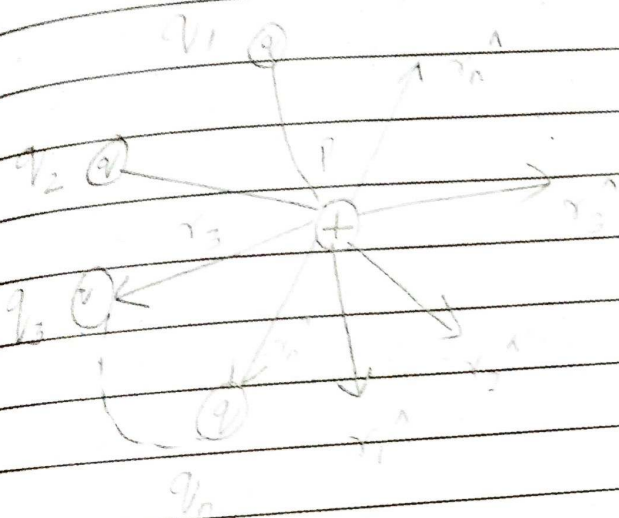
7) Put the value of $\vec{F}$ in eq 1

$$E = \frac{1}{4\pi\epsilon_0 r^2} \frac{q \cdot q}{q} r^{\wedge}$$

$$E = \frac{1}{4\pi\epsilon_0 r^2} \times q r^{\wedge}$$

This gives the electric field intensity due to charge

Electric Field Intensity Due to n-charge



① Consider n-point charges

$q = q_1 + q_2 + q_3 + \dots + q_n$
 placed at a distance of $r_1 + r_2 + r_3 + \dots + r_n$ from point P

② To determine the electric field intensity at point 'P' due to these charges at point P. Put a charge 'q' over there.

③ The electric field intensity due to each charge can be determined as;

$$\vec{F}_1 = \frac{1}{4\pi\epsilon_0} \frac{q_1}{r_1^2} \hat{r}_1$$

$$\vec{F}_2 = \frac{1}{4\pi\epsilon_0} \frac{q_2}{r_2^2} \hat{r}_2$$

$$\vec{F}_n = \frac{1}{4\pi\epsilon_0} \frac{q_n}{r_n^2} \hat{r}_n$$

Total E.F

$$E_{\text{TOTAL}} = \vec{E}_1 + \vec{E}_2 + \dots + \vec{E}_n$$

$$E_{\text{TOTAL}} = \frac{1}{4\pi\epsilon_0} \left(\frac{q_1}{r_1^2} \hat{r}_1 + \frac{q_2}{r_2^2} \hat{r}_2 + \dots + \frac{q_n}{r_n^2} \hat{r}_n \right)$$

$$E_{\text{TOTAL}} = \frac{1}{4\pi\epsilon_0} \left[\text{Sum of } \frac{q's}{r^2} \right]$$

$$E_{\text{TOTAL}} = \frac{1}{4\pi\epsilon_0} \sum_{n=1}^n \frac{q_n}{r_n^2} \hat{r}_n$$

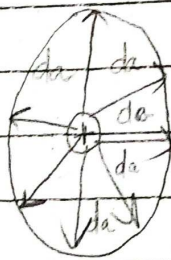
This gives the total electric field intensity due to n-charges.

ELECTRIC FLUX

The number of electric lines passing through the sheet.

The dot product electric field vector and area da

Electric Flux due to a charge when charge is placed at the centre of the spheres



$$\phi_1 = E \cdot da_1$$

$$\phi_2 = E \cdot da_2$$

$$\phi_3 = E \cdot da_3$$

$$\phi_n = E \cdot da_n$$

$$\begin{aligned}
 \phi_{\text{total}} &= E \cdot da_1 + E da_2 + \dots E da_n \\
 &= E [da_1 + da_2 + \dots da_n] \\
 &= E \cdot [\text{Total Area of sphere}] \\
 &= \frac{1}{4\pi\epsilon_0} \cdot \frac{q}{r^2} [4\pi r^2]
 \end{aligned}$$

$$\phi_{\text{total}} = \frac{q_{\text{total}}}{\epsilon_0}$$

SOLID ANGLES

Diagram?

The angle subtended by the two dimensional to any point is known as **Solid Angle**

Mathematically,

$$\text{Solid Angle} = \frac{\text{Vector Area}}{r^2}$$

$$d\Omega = \frac{da}{r^2} r^1$$

$$d\Omega = \frac{|da| |\hat{n}| |\hat{r}| \cos\theta}{r^2} \quad \text{where } \theta = 0$$

$$d\Omega = \frac{da}{r^2}$$

integrating,

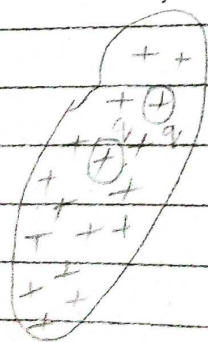
$$\int d\Omega = \frac{1}{r^2} \int da$$

$$\Omega = \frac{1}{r^2} [4\pi r^2]$$

$$\Omega = 4\pi \text{ steradian}$$

GAUSS'S LAW (Integral Form)
OR

FLUX DUE TO n -CHARGES



Consider n -point charges $q_1, q_2, q_3, \dots, q_n$ enclosed in surface.

To determine electric flux, due to these charges we first determine the electric flux, suppose we place at the centre of surface.

we know that :

$$\phi = \int E \cdot da$$

$$\therefore \vec{E} = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2} \hat{r}$$

$$\phi = \int \frac{q}{4\pi\epsilon_0 r^2} \hat{r} \cdot da$$

$$\frac{q}{4\pi\epsilon_0} \int \frac{da}{r^2} \cdot \hat{r}$$

$$\frac{q}{4\pi\epsilon_0} \int \frac{da \hat{n} \cdot \hat{r}}{r^2}$$

$$\frac{q}{4\pi\epsilon_0} \int \frac{da |\hat{n}| |\hat{r}| \cos\theta}{r^2}$$

$$\frac{q}{4\pi\epsilon_0} \times 4\pi$$

$$\frac{q}{\epsilon_0}$$

$$\Rightarrow \phi_1 = \frac{q_1}{\epsilon_0}$$

$$\Rightarrow \phi_2 = \frac{q_2}{\epsilon_0}$$

$$\Rightarrow \phi_3 = \frac{q_3}{\epsilon_0}$$

$$\Rightarrow \phi_n = \frac{q_n}{\epsilon_0}$$

Total Flux

$$\begin{aligned} \phi_{\text{total}} &= \phi_1 + \phi_2 + \phi_3 + \dots + \phi_n \\ &= \frac{q_1}{\epsilon_0} + \frac{q_2}{\epsilon_0} + \frac{q_3}{\epsilon_0} + \dots + \frac{q_n}{\epsilon_0} \end{aligned}$$

$$= \frac{1}{\epsilon_0} (q_1 + q_2 + \dots + q_n)$$

$$\phi_{\text{total}} = \frac{1}{\epsilon_0} \times q_{\text{total}}$$

Integral form of Gauss's Law, the flux and electric field through any enclosed surface is equal to

$$\phi_{\text{total}} = \frac{1}{\epsilon_0} \times q_{\text{total}}$$

GAUSS'S LAW OR

(Differential form)

FLUX DUE TO n -CHARGES



Point 1 same as previous Integral Form

From integral form of Gauss's Law

$$\phi = \frac{q_{\text{total}}}{\epsilon_0} \rightarrow \textcircled{i}$$

and

$$\phi = \int E \cdot da \rightarrow \textcircled{ii}$$

From eq (i)

$$\phi = \frac{q_{\text{total}}}{\epsilon_0}$$

$$q_{\text{total}} = q_v = \int \rho dv$$

so,

$$\phi = \frac{\int \rho dv}{\epsilon_0} \rightarrow \textcircled{iii}$$

equating (ii) and (iii)

$$\int E \cdot da = \frac{\rho}{\epsilon_0} \int dv$$

Transforming the Surface Integral to volume integral by Gauss's Divergence Theorem

$$\Rightarrow \int E \cdot da = \int \text{Div } E \, dv$$

so we can write

$$\Rightarrow \int \text{Div } E \, dv = \frac{\rho}{\epsilon_0} \int dv$$

$$\Rightarrow \text{Div } E \int dv = \frac{\rho}{\epsilon_0} \int dv$$

$$\Rightarrow \text{Div } E [v] = \frac{\rho}{\epsilon_0} [v]$$

$$\Rightarrow \text{Div } E = \frac{\rho}{\epsilon_0}$$

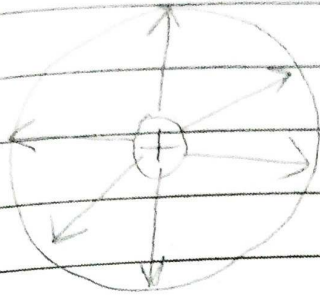
$$\nabla \cdot E = \frac{\rho}{\epsilon_0} \rightarrow \text{Differential form}$$

$$\left(\frac{\partial}{\partial x} i + \frac{\partial}{\partial y} j + \frac{\partial}{\partial z} k \right) \cdot (E_x i + E_y j + E_z k) = \frac{\rho}{\epsilon_0}$$

$$\frac{\partial E_x}{\partial x} + \frac{\partial E_y}{\partial y} + \frac{\partial E_z}{\partial z} = \frac{\rho}{\epsilon_0}$$

$\rightarrow$ Differential form of Gauss's Law

COLUMB'S LAW FROM GAUSS'S LAW



$$\oint E \cdot da$$

$$\Rightarrow \phi = E \cdot da - (i)$$

$$E da \cos\theta = E da$$

$$\Rightarrow \phi = \frac{q}{\epsilon_0} - (ii)$$

$\Rightarrow \phi$ also equal to $\oint E \cdot da$

equate eq (i) and (ii)

$$\frac{q}{\epsilon_0} = \oint E \cdot da$$

$$q = \epsilon_0 E \int da$$

$$\text{or } \int da = 4\pi r^2$$

$$\Rightarrow q = \epsilon_0 E (\text{Area of the sphere})$$

$$\Rightarrow q = \epsilon_0 E \times 4\pi r^2$$

$$q E = \frac{q}{4\pi \epsilon_0 r^2}$$

We know that

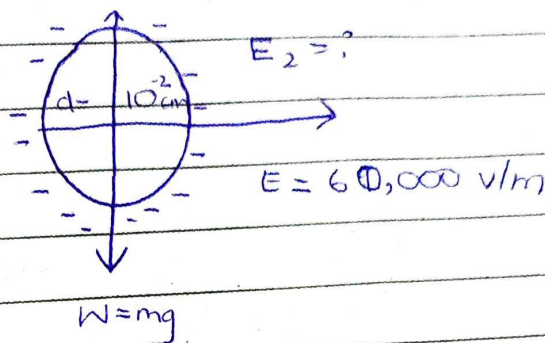
$$F = qE$$

Put the value of E

$$F = q \times \frac{q}{4\pi \epsilon_0 r^2}$$

$$F = \frac{1}{4\pi \epsilon_0} \cdot \frac{qq}{r^2}$$

Q2 A water droplet 10^{-2} cm in diameters carries a negative charge such that the magnitude of the electric field at its surface is 60000 V/m. How strong a vertical field be required to keep the droplet from falling.



$$d = 10^{-2} \text{ cm}$$

$$r = 5 \times 10^{-5} \text{ m}$$

$$E = 60,000$$

For charge 'q'

$$E = \frac{k \cdot q}{r^2}$$

$$q = \frac{E r^2}{k}$$

$$q = 1.6 \times 10^{-14} \text{ C}$$

Now,

$$F_e = F_w$$

$$q E_2 = mg$$

$$E_2 = \frac{mg}{q}$$

$$\text{∴ } e = \frac{m}{V} \Rightarrow m = eV$$

FOR VOLUME 'V'

$$\text{∴ } V = \frac{4}{3} \pi r^3$$

$$V = \frac{4}{3} (3.142) (5 \times 10^{-5})^3$$

$$V = 5.3 \times 10^{-13} \text{ m}^3$$

FOR MASS

$$\text{∴ } m = e \times V$$

$$m = ~~1.5~~ 1 \times 5.3 \times 10^{-13}$$

$$m = 5.3 \times 10^{-13} \text{ kg}$$

$$\text{∴ } E_2 = \frac{mg}{q} \Rightarrow \frac{5.3 \times 10^{-13} \times 9.8}{1.6 \times 10^{-4}}$$

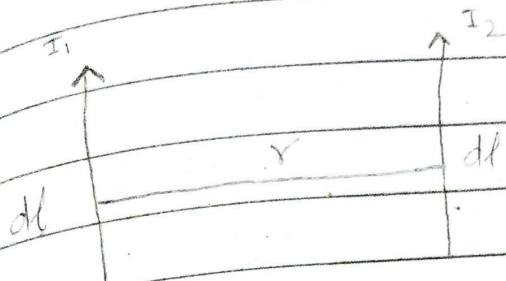
$$E_2 = 324.6 \text{ V/m}$$

$$\text{∴ } E = E_1 + E_2$$

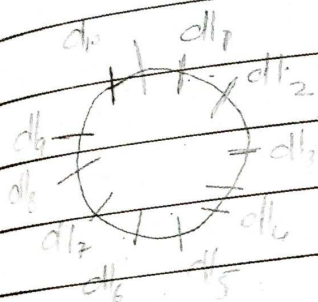
$$E = 60000 + 324.6$$

$$E = 60324.6 \text{ V/m}$$

Force B/w Two Current Carrying Conductor



$$F = \frac{\mu_0}{2\pi} \frac{dl I_1 I_2}{r}$$



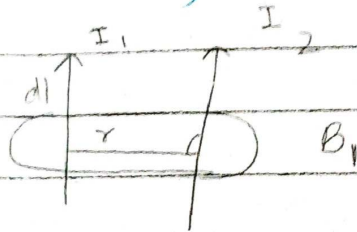
$$\begin{aligned} \oint &= 2\pi \\ &= \text{orbital circumference} \\ &= dl_1 + dl_2 + \dots \\ &= \text{sum of } dl\text{'s} \\ &= \int dl \end{aligned}$$

AMPERE'S LAW

(Integral Form)

The line integral of the product of tangential magnetic field 'B' and the path element 'dl' equal to the ~~the~~ μ_0 times the current enclosed by the circle.

$$\oint B \cdot dl = \mu_0 \times I$$



1) Consider a conductor of ~~current~~ length 'dl' carrying ~~current~~ ^{of current} I_1 Ampere (~~A~~), surrounded by the magnetic field B_1 .

2) Let another current carrying conductor of some length replaced in the magnetic field 'B'.

3) The force exerted by the magnetic field 'B' of the current carrying conductor I_1 , Ampere of current is given by ;

$$F = I_2 (dl \times B_1)$$

$$F = I_2 dl B_1 \sin \theta$$

$$F = I_2 dl B_1 \sin 90^\circ \quad \because \theta = 90^\circ$$

$$F = I_2 dl B_1 \quad \rightarrow \text{eq. i}$$

The force b/w two current carrying conductor is given by;

$$F = \frac{\mu_0}{2\pi} \cdot \frac{dl I_1 I_2}{r} \quad \rightarrow \text{eq. ii}$$

equating (i) and (ii)

$$I_2 dl B_1 = \frac{\mu_0}{2\pi} \cdot \frac{dl I_1 I_2}{r}$$

$$\Rightarrow B_1 = \frac{\mu_0}{2\pi} \cdot \frac{I_1}{r}$$

OR

$$\Rightarrow B = \frac{\mu_0 \cdot I}{2\pi r}$$

$$\Rightarrow B \cdot 2\pi r = \mu_0 I$$

$$\Rightarrow B \cdot (2\pi r) = \mu_0 I$$

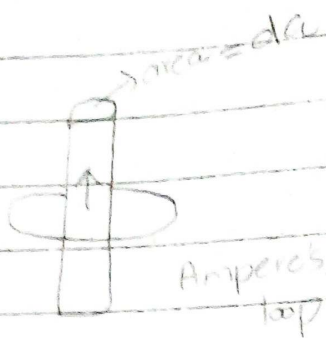
$$\Rightarrow B (\text{orbital circumference}) = \mu_0 I$$

$$\Rightarrow B \cdot (dl_1 + dl_2 + \dots) = \mu_0 I$$

$$\Rightarrow \int B \cdot dl = \mu_0 I$$

AMPERE'S Law

Differential Form



The integral form of Ampere's Law can be written as

$$\oint \mathbf{B} \cdot d\mathbf{l} = \mu_0 I \rightarrow i$$

ii) we know that
current density = $\frac{\text{current}}{\text{area}}$

$$j = \frac{I}{da}$$

$$I = j da \quad \text{or}$$

$$I = j \int da$$

Put value of I in (i)

$$\oint \mathbf{B} \cdot d\mathbf{l} = \mu_0 \int j \cdot da$$

Transforming by Stokes theorem

$$\oint \mathbf{B} \cdot d\mathbf{l} = \int \text{curl } \mathbf{B} \cdot da$$

so above expression

$$\int \text{curl } \mathbf{B} \cdot da = \mu_0 \int j \cdot da$$

$$\text{curl } \mathbf{B} [a] = \mu_0 j [a]$$

$$\text{curl } \mathbf{B} = \mu_0 j$$

$$\nabla \times \mathbf{B} = \mu_0 j$$

show that:

B is produced by J
magnetic field ' g ' is produced by current
density J .

B is circular

Q) A three level laser produce a light of wavelength 550 nm, at what temperature the ratio of population is half (N_2/N_1)

$$\lambda = 550 \text{ nm}$$

$$\frac{N_2}{N_1} = \frac{N_0 e^{-E_2/KT}}{N_0 e^{-E_1/KT}} = \frac{1}{2}$$

$$e^{-E_2 - E_1 / KT} = \frac{1}{2}$$

$$\frac{1}{2} = e^{-\Delta E / KT}$$

$$\frac{1}{2} = \frac{c}{\lambda} = \frac{3 \times 10^8}{550 \times 10^{-9}}$$

taking log

$$\ln \frac{1}{2} = -\frac{\Delta E}{KT} = \ln \frac{1}{2}$$

$$\ln \frac{1}{2} = -\frac{\Delta E}{KT}$$

$$T = 37.15^\circ \text{ K}$$

N_2/N_1

✓ Calculate the population ratio of helium neon laser that produces the light of wavelength $6678 \text{ \AA}$ at 300 K and 1 eV

$$\frac{N_2}{N_1} = e^{-80}$$

✓ A laser source has a power output of 10^{-3} watt . Calculate the no. of photon emitted per second given the wavelength of laser light is 612.8 nm

$$\rightarrow 3.48 \times 10^{-15} \rightarrow \text{no. of photon}$$

✓ A. Krpton cathode whose threshold wavelength is $2300 \text{ \AA}$ is irradiated by ultraviolet light of wavelength $1800 \text{ \AA}$. Calculate max kinetic photo-electrons emitted. The work function for Krpton is 5.4 eV .

$$\lambda_0 = 2300 \text{ \AA}$$

$$\lambda = 1800 \text{ \AA}$$

1) work function

$$\phi_0 = h\nu_0$$

$$\phi_0 = \frac{hc}{\lambda_0}$$

$$\phi_0 = \frac{6.625 \times 10^{-34} \times 3 \times 10^8}{2300 \times 10^{-10}}$$

$$\phi_0 = 8.6 \times 10^{-19} \text{ J}$$

$$\phi_0 = \frac{8.6 \times 10^{-19}}{1.6 \times 10^{-19}} = 5.4 \text{ eV}$$

$$\textcircled{1} \quad h\nu = \phi_0 + K.E$$

$$K.E = h\nu - \phi_0$$

$$= \frac{hc}{\lambda} - \phi_0$$

$$K.E = 1.5 \text{ eV} \quad 6.9$$

Date _____

Q. When violet light of $4100 \text{ \AA}$ is incident on the cathode of the photocell of retarding/stopping potential of 0.4 V is required to stop the emission of light.
1) Calculate